



DATA INTEGRITY REVIEW

Finding the bots in the caregiver survey data

What we can prove, what we cannot, and what we do next

Early Social Development Lab | Institute for Mind and Brain | 19 August 2026

Where we are

One survey project is clean. The other was flooded.

1,956

records screened

177

in the verified project

1,779

in the flooded project

31

people we can vouch for

Everything in this deck comes back to that last number.

We proved the humans. We only guessed the bots.

Every bot number in the report carries that difference.

31 real caregivers

Matched to email addresses we recognise.
This is proof.

111 suspected bots

Flagged because the behaviour looked wrong.
This is a judgement call.

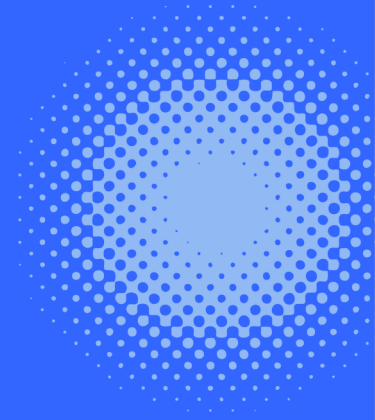
- Email addresses were removed before the file left the lab. The analyst was not on our protocol.
- That took away the signal most of us think is the strongest one we have.

01

PART ONE

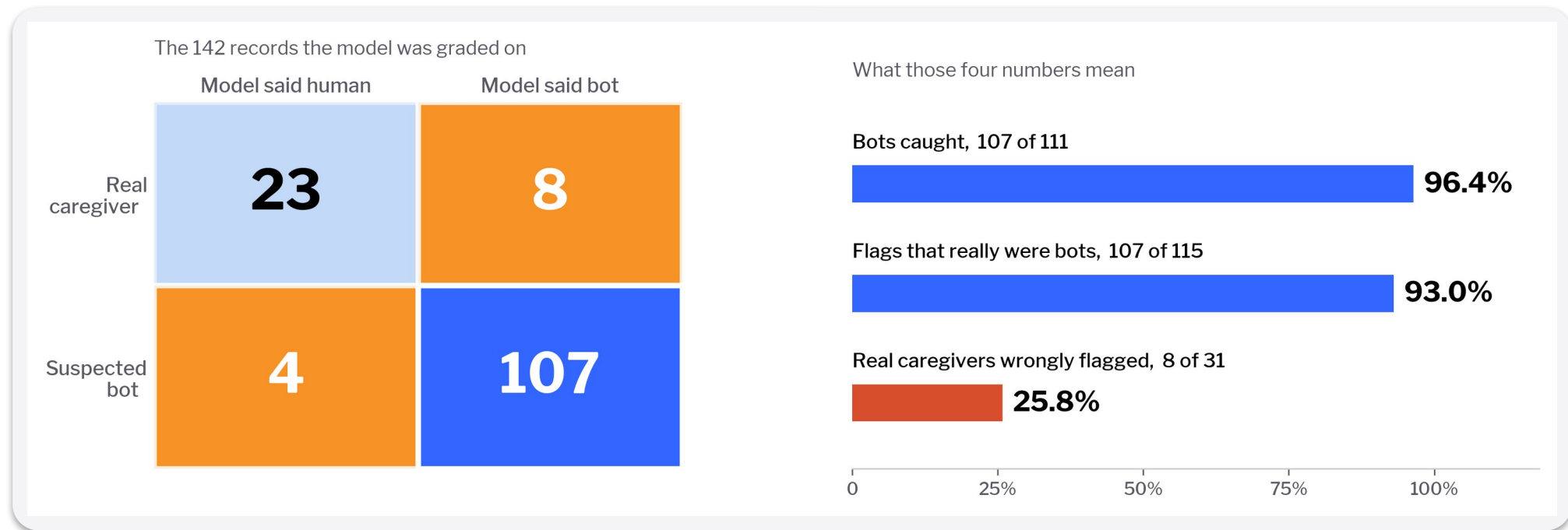
The model we were sent

What it does, and what it costs



It tells the two piles apart

Give it one real caregiver and one suspected bot. It puts them in the right order 94 times out of 100.



- It uses 21 simple signals. Mostly how fast people answered, what they clicked and how much they typed.
- It was graded on records it had never seen, so the score is not just memory.

Eight of our 31 real caregivers were called bots

This is the number the room should argue about.

8 of 31

real caregivers the model
wrongly flagged as bots

- That is one in four of the people we can actually vouch for.
- If a flag ever stops a payment, those families pay for our convenience.
- Move the cut-off and this number moves. The team should pick it, not the model.

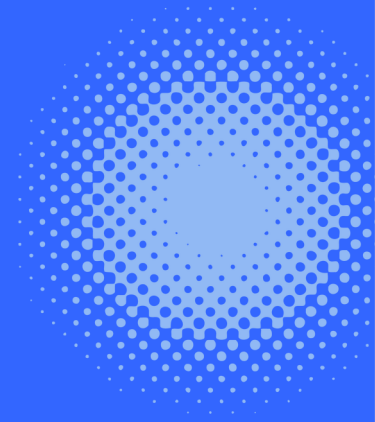
The model catches 96 of every 100 suspected bots. Both numbers move together.

02

PART TWO

Your five questions

Answered one at a time



QUESTION 1

Where the signals came from

Two sources, and one thing worth fixing.

From the research

- How fast people answer
- Whether they skip questions
- Picking the same answer all the way down a scale

From this dataset

- Emotion ratings
- The values scale answers
- How much people typed

Worth fixing

- The signals were picked using the same 142 records the model was then graded on
- That flatters the grade
- Split the labels once, before picking

QUESTION 2

Why the same answer down a scale looks small

Both statements in the report are true. They come from two different checks.

- On its own, picking the same answer down a whole scale does separate the two groups.
- Inside the model it barely registers, because the pacing signals already say the same thing.
- It is not a weak signal. It is a repeated one. Report it as backup, not as the headline.

CHECK ONE, ON ITS OWN

Top two signals

What the report summary quotes

1. How long the survey took
2. Same answer down a scale

Both lists are right.
They ask different
questions.

On its own, picking the same answer down a scale does separate the two groups. Next to the pacing signals, it has almost nothing left to add.

CHECK TWO, ALL SIGNALS AT ONCE

Top signals

What the importance chart shows

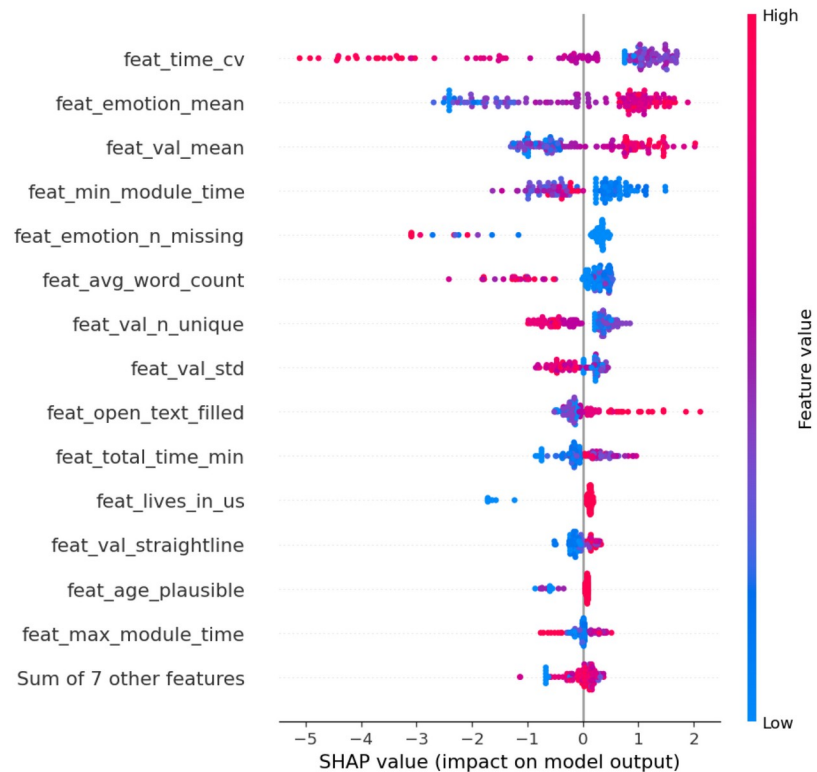
1. Uneven pace across sections
2. Emotion ratings
3. Values scale answers
4. Fastest single section
5. Skipped emotion ratings
6. Words typed
- ...

10. How long the survey took
12. Same answer down a scale

QUESTION 3

How to read the importance chart

Three things, and the chart opens up.



1 One dot is one person.

Each of the 142 records appears once on every row.

2 Left or right is the nudge.

Right of the line nudged that person toward bot.
Left nudged them toward human.

3 Colour is that person's score.

Pink is a high value of the signal on that row.
Blue is a low one. Colour is not the answer.

WHAT THE ROW NAMES MEAN

time_cv	How uneven the pace was
emotion_mean	Average emotion rating
val_mean	Average values scale answer
min_module_time	Fastest single section
val_straightline	Same answer down the scale

Read the top row this way. Pink sits far left, so uneven pacing looked human.

QUESTION 4

What about all the other records

The report gives no full count, and that was the right call.

- Only 142 of the 1,637 records carry a label. The rest were never scored in the report.
- At a cut-off that misfiles one in four real people, a full count would be a guess.
- Our own screen covers every record with fixed rules instead.

WHAT OUR SCREEN SAYS

1,956

records screened, all of them

223

pass every single rule

86.4%

of the verified project passes

3.9%

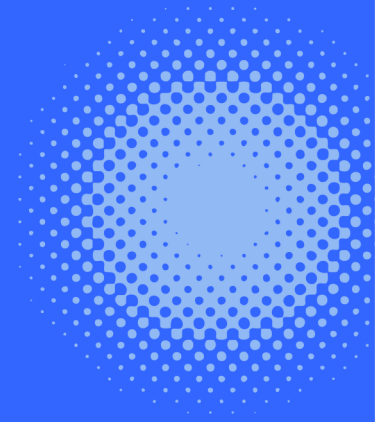
of the flooded project passes

03

PART THREE

What we run instead

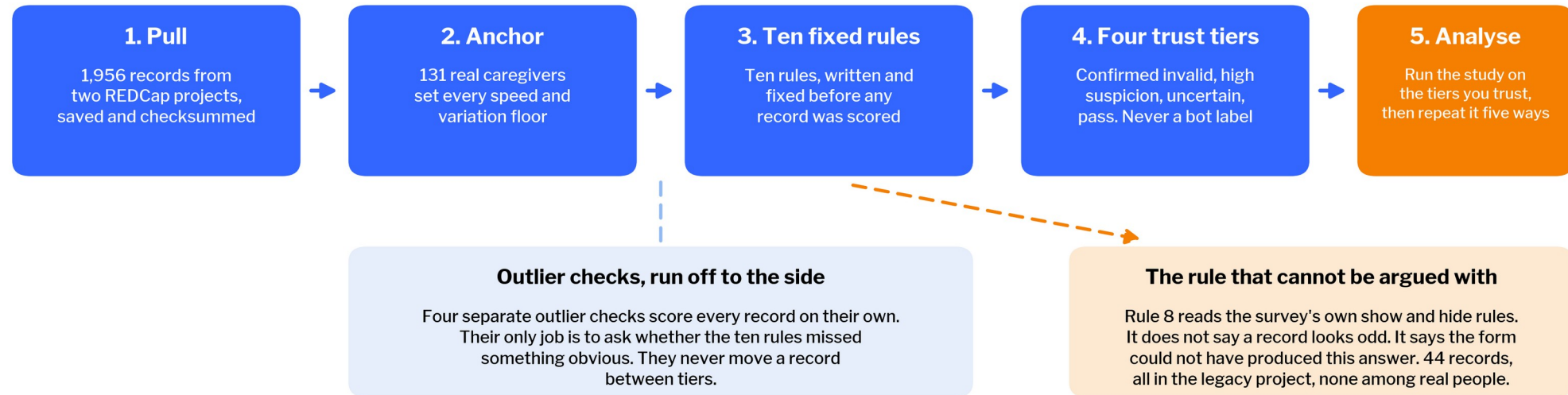
Rules first, models second



Rules decide. Models only check the rules.

No family is ever removed from a study by a score alone.

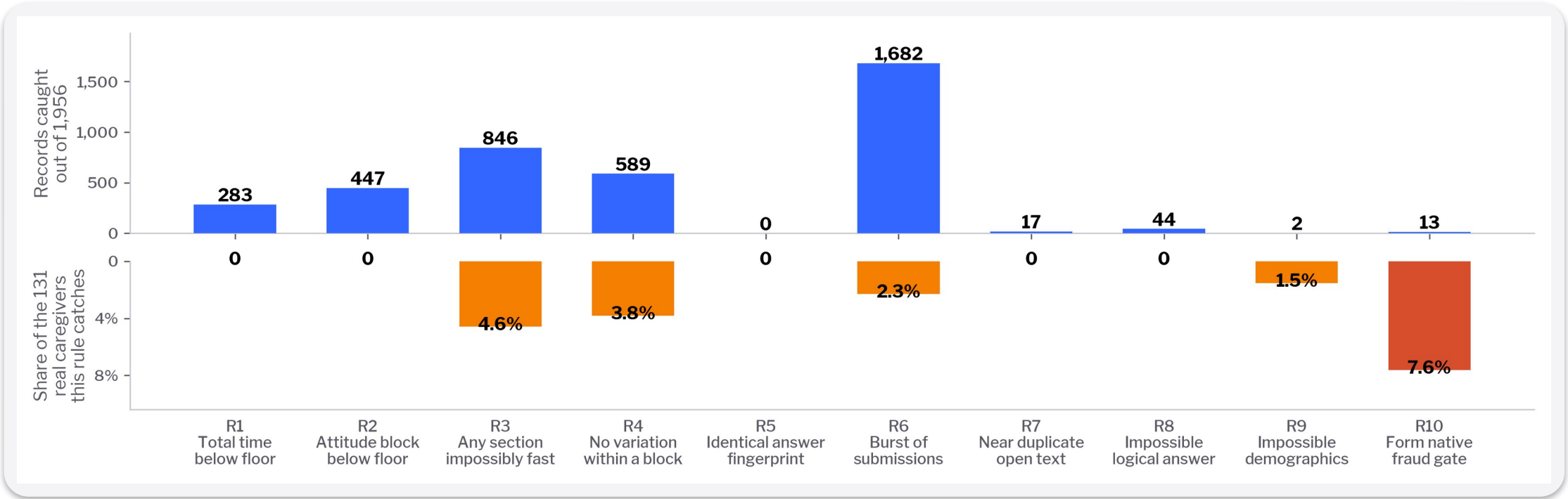
THE SCREEN RUNS LEFT TO RIGHT. EVERY GATE IS A RULE A PERSON CAN READ.



- The speed floors come from 131 real caregivers, so they are what real families actually did.
- The rules were fixed before any record was scored, so nobody tuned them to get an answer.

Every rule reports how many real people it wrongly catches

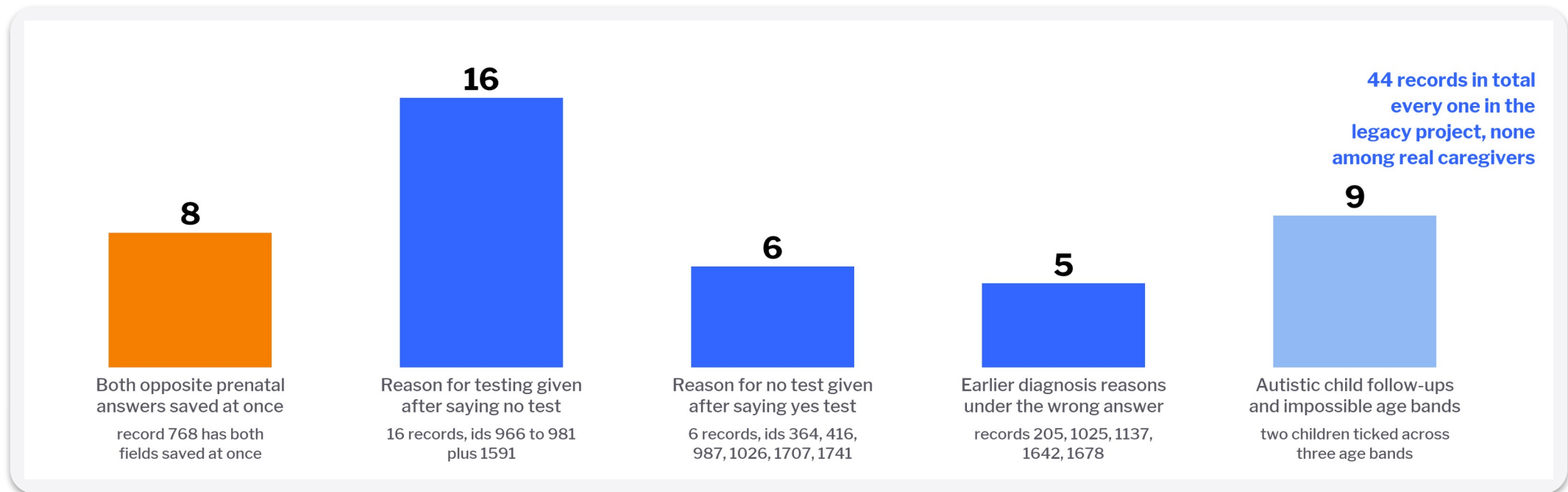
Five of the ten catch none of our 131 real caregivers.



The rules that cost nothing are the ones built on impossible, not on unusual.

Answers the form could not have produced

The survey hides questions based on earlier answers. These records answered questions they were never shown.

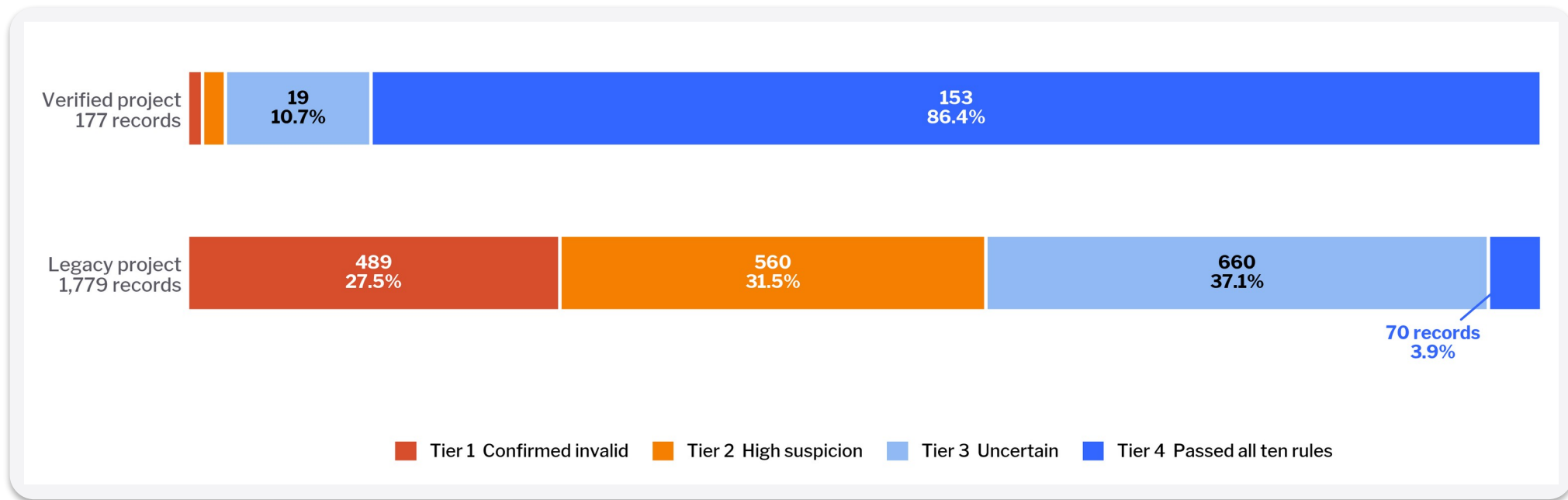


- 44 records break one of these checks. Every one of them is in the flooded project.

- None of our 131 real caregivers breaks a single one.

A ladder, not a label

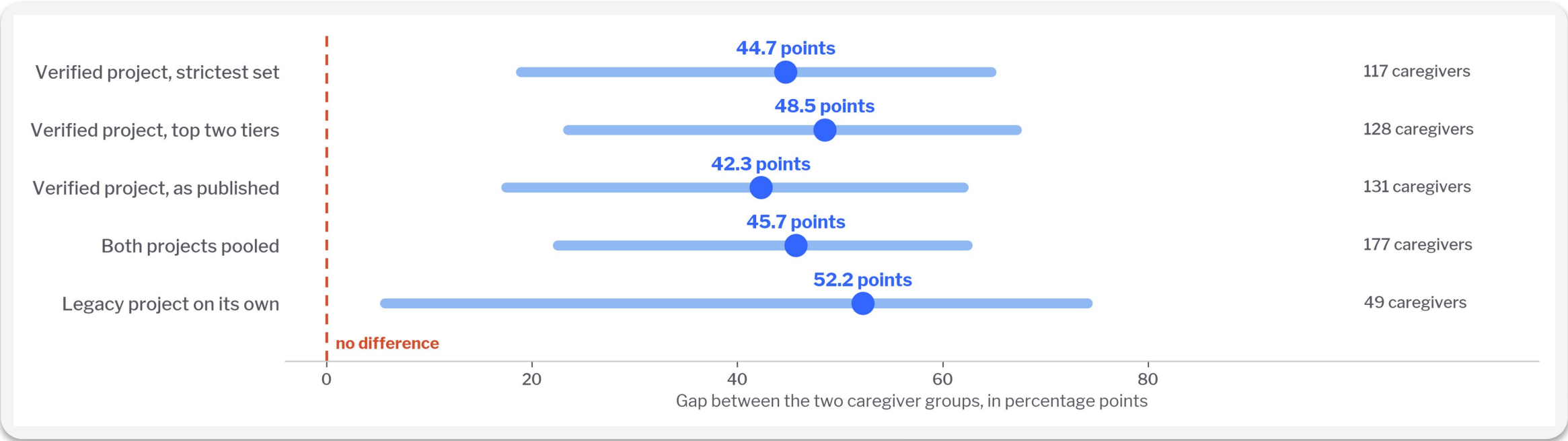
No record is ever called a bot. Each one gets a step, and we say which steps we used.



- Tier 1 broke a hard rule. Tier 4 passed all ten. The middle is where judgement lives.
- That gap between the two projects is itself the finding.

We reran the whole study five times

Each run kept a different set of records. The answer never came near zero.



This is what lets us say the finding survives the cleaning choice, rather than depending on it.

QUESTION 5

How to build a survey bots find hard to fill in

Most of this is winnable before a single record arrives.

When you design it

- Add a hidden field. Anything that fills it is a machine.
- Make hidden questions impossible to answer, not just hidden.
- Put two backwards worded items in every scale.

While it collects

- Timestamp every section, not just the submission.
- Limit how many entries one session can send.
- Pay after screening, not before.

Before you trust it

- Run the ten rules nightly, not once at the end.
- Report a tier for every record, never a bot label.
- Publish how many real people each rule catches.

Four decisions

None of these need new modelling. They need the room to agree.

1

Keep the verified project as primary

Treat the flooded one as a cautious second opinion.

2

Report a tier for every record

Publish no bot count and no bot percentage.

3

Pick the cut-off together

Tell us what a wrongly excluded family costs. The rest follows.

4

Ship the stronger form rule

Then rerun the screen across both projects.

THE ONE THING WE NEED BACK

Bring the email address back inside the analysis scope

- Everyone who has looked at this data thinks the email pattern is our strongest signal.
- It is also the one signal the outside analysis was never allowed to use.
- Bringing it inside our own protocol costs us nothing scientifically.

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